Technological Progress and Power

Technological progress has a marked impact on power in society. This is not determined by the technology itself; the effects of progress on power are based on the social environment that technology emerges into. Mokyr (1990) defines technological progress as "any change in the application of information to the production process in such a way as to increase efficiency", either lowering the resource cost of existing outputs, or creating entirely new/better products (pg. 6). It creates a "free lunch" for society - an "increase in output that is not commensurate with the increase in effort and cost necessary to bring it about" (Mokyr 1990, pg. 3). Mokyr (1990) divides progress into microinventions, small incremental developments; and macroinventions, radical new ideas that cause disruptions (p. 13). The disruptive process of integrating new technology "results in winners and losers and often affects established interests… It always generates both positive and negative outcomes" (Dalvit et al. 2023, p. VII). The distribution of 'free lunch' and these positive or negative outcomes is a question of power. Lukes (2005) identifies three dimensions of power: decision making, agenda-setting, and the shaping of preferences and perceptions (p. 29). Clegg (2023) organises power into three circuits; episodic, dispositional, and facilitative (p. 28). While technological progress can potentially distribute power by democratising access to knowledge, it has concentrated power by creating infrastructure dependencies that enable monopolistic control by companies. This is compounded by already-powerful agents using their influence to set the agenda for future progress and shape public preferences on new technology in ways that maintain their dominance. This concentration of power is not determined by technological progress alone but is a product of the social and institutional frameworks into which technologies emerge and can be mitigated through deliberate policy intervention.

Technological progress can distribute power by expanding access to knowledge, allowing more people to participate in productive industries and idea sharing. This is an example of facilitative power. Clegg (2023) describes facilitative power as productive capacity; "the empowerment and disempowerment of agencies' capabilities" (p. 233) and the "ability to achieve goals" (Clegg 2023, p. 6). Access to information enables growth in capacity, distributing facilitative power. Knowledge can be distributive because it is "non-rival in use: the same idea… may be used repeatedly and concurrently by many people" (David & Foray 2003, p. 38). An historical example of a macroinvention that distributed power is the printing press, which radically increased access to knowledge; "…in the 50 years following the invention more books were

produced than in the preceding thousand years" (Mokyr 1990, p. 49). A contemporary case is discussed by Liu et al. (2024), who examined 32,000 households in India and found that higher computer ownership in a community helped "disadvantaged castes [to] overcome low-income mobility" (p. 1425). Even when households did not own computers, their mobility was enhanced by proximity through regional knowledge spillover (Liu et al. 2024, p. 1442). Knowledge spillover is where "…external benefits from the creation of knowledge… accrue to parties other than the creator" (Agarwal et al. 2010, p. 271) and can lead to innovation that synthesises information from multiple fields. By reducing barriers to knowledge sharing, technological progress enables the distribution of facilitative power. However, while knowledge is non-rival in nature, the infrastructure through which it is implemented is subject to intense concentration.

Technological progress concentrates power by enabling the creation of monopolies that build dependency. Vasudevan (2022) argues that natural monopolies arise "as the outcome of technology and the production process", due to the "significance of barriers to entry created by economies of scale and large sunk costs" (p. 1272). This is particularly true of macroinventions that require large-scale infrastructure construction, like railroads in the nineteenth century and digital platforms today. Vasudevan (2022) argues the dominance of contemporary tech corporations echoes the monopolies of nineteenth century railroad companies (p. 1280). She highlights that railroad monopolies were not merely "natural" but also "artificial"; corporations deliberately pursued monopolisation and consolidation (Vasudevan 2022, p. 1280). For example, "six railroad groups… used their control of railroad tracks to obstruct coal haulage by independent coal companies who depended on their infrastructure" (Vasudevan 2022, p. 1280). This is a clear example of episodic power, which Clegg (2023) defines as direct "power over" others, "where A exercises power over B by getting B to do something that they would not otherwise do" (p. 20). Railroad companies frequently merged into trusts, concentrating their episodic power and giving them "outsize[d] power… over the whole economy" (Vasudevan 2022, p. 1280). Companies such as Amazon, Google and Facebook embody the same monopolistic characteristics, with even greater scope (Vasudevan 2022, p. 1280). Integrated digital systems have enabled unprecedented monopolies, that are not reliant on physical infrastructure and can readily expand across national borders (Vasudevan 2022, p. 1280). Digital platforms monetise intangible capital, improving their services by using user data (Vasudevan 2022, p. 1280). This creates network effects: feedback loops where newly acquired services bring in users, additional users "increases the overall value of the platform to other

users", generating more data that "can be harnessed to further expand the reach of the platform" (Vasudevan 2022, p. 1282). The resultant concentration of services has led to platforms "simultaneously exercising monopoly power over users and monopsony power over suppliers" (Fatmawati et al. 2025, p. 559). Technological progress concentrates power when new opportunities for monopolisation and induced dependency are created.

Beyond the episodic power enabled by infrastructure dependencies, corporations exercise power by shaping the direction of progress itself. Lukes (2005) identifies this as second dimension power, control over agenda (p. 25). Through the "mobilisation of bias", certain fields of research are prioritised and "organised into politics while others are organised out" (Schattschneider, cited in Lukes 2005, p. 20). Recent innovation has been steered heavily towards automation and AI, with technology companies spending hundreds of millions of dollars training each new model (Canboy & Khlif 2025, 109550-6). Acemoglu and Restrepo (2018) highlight the "singular focus of the research… community on automation, at the expense of other types of technologies…" (24196-3). This is corroborated by Dani Rodrik, who notes "the ethos in Silicon Valley and the innovation community… favours labour-replacing technologies" (cited in Dalvit et al. 2023, p. 32). Korinek et al. (2021) advocate for "actively steering technological progress so that it is more labour-using" (p. 32) however, this shift is prevented by the second-dimensional power of corporations. By funnelling vast resources into data centres and innovation geared towards labour-replacement, investment into alternative fields of research becomes disincentivised; they are 'organised out' of research agendas. The dominance of these firms is further insulated from resistance through their exercise of Lukes' (2005) third dimension of power to manufacture consent from users by "shaping their perceptions, cognitions and preferences in such a way that they accept their role" (p. 28). Culpepper and Thelen (2020) argue that firms that reach certain economic scales become integrated into the daily lives of users, conferring a "tight, even intimate, connection" (p. 290). Firms cultivate "privileged alliances" with consumers, who become dependent on their services and are the most "formidable source of opposition to regulation that threatens the convenience provided by these platforms" (Culpepper & Thelen 2020, p. 290). By setting the agenda of innovation and shaping their users' preferences, corporations ensure that the "free lunch" of technological progress flows to capital instead of labour.

Concentration of power is not the inevitable outcome of technological progress. Rather, current social and institutional frameworks lend themselves towards inequitable divisions of surplus. This represents the dispositional circuit of power, which Clegg (2023) defines as "rules that fix

relations of meaning and membership" (p. 28). Clegg (2023) uses this to explain how domination by certain groups becomes institutionalised in society (p. 223). Existing rules disproportionately favour corporations, due to their greater dispositional power, so technological progress reinforces their dominance rather than disrupting it. This increases market concentration "with a few… 'superstar' firms outcompeting smaller, less productive ones" and has created a "paradox of rising inequality and stagnant productivity during a period of rapid technological progress" due to the stifling of innovation (Dalvit et al. 2023, p. VII). This mirrors a historical "trap" discussed by Mokyr (1990), Cardwell's Law: "no nation has been (technologically) very creative for more than an historically short period" (p. 207). Mokyr (1990) describes how powerful groups would "stifle technological progress to protect their monopolistic position" (p. 77), arguing that Europe escaped technological regression because its political fragmentation created competition (Mokyr 1990, p. 207). 'Superstar' firms face little competition, risking technological stagnation where innovation is directed only towards maintaining power. To avoid this, regulatory intervention is required. Anti-trust laws were enforced on railroad monopolies in the nineteenth/early twentieth centuries to divide their economic and political clout (Vasudevan 2022, p. 1279). Enforcement of anti-trust law is not the only way policy can distribute the benefits of progress more equally. Dalvit et al. (2023) discuss how Estonia has "emerged as one of the most technologically advanced societies in the world" (p. 86). By implementing technology into its education system, promoting entrepreneurial activity, and digitising government services, Estonia "made its tech sector an engine of growth and job creation, that critically contributes to its shared prosperity" (Dalvit et al. 2023, p. 86). This highlights how technological progress can have positive, distributive effects on power if the dispositional power of corporations is addressed through strong institutional intervention.

Whether technological progress distributes or concentrates power is not a matter of technological determinism, but a reflection of the social and institutional frameworks into which new technologies arrive. While innovations in knowledge accessibility can distribute power by improving access to information and facilitating knowledge spillovers, this is frequently overshadowed. Already-powerful agents use advantaged positions to create dependencies and steer progress in their preferred direction, using technological advances to further concentrate power. This can be mitigated through policy interventions, ensuring that the "free lunch" of technological progress can be enjoyed by everyone.

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